Latest updates: https://dl.acm.org/doi/10.1145/3392063.3394429

RESEARCH-ARTICLE

LifeMosaic: co-design of a personal informatics tool for youth

KYRILL POTAPOV, University College London, London, U.K.

PAUL MARSHALL, University of Bristol, Bristol, U.K.

Open Access Support provided by:

University College London

University of Bristol

PDF Download
3392063.3394429.pdf
25 March 2026
Total Citations: 11
Total Downloads: 769

Published: 21 June 2020

Citation in BibTeX format

IDC '20: Interaction Design and Children
June 21 - 24, 2020
London, United Kingdom

Conference Sponsors:
SIGCHI

IDC '20: Proceedings of the Interaction Design and Children Conference (June 2020)
https://doi.org/10.1145/3392063.3394429
ISBN: 9781450379816

LifeMosaic: Co-design of a Personal Informatics Tool for Youth

Kyrill Potapov
UCLIC, London, UK
kyrill.potapov.16@ucl.ac.uk

Paul Marshall
University of Bristol, UK
p.marshall@bristol.ac.uk

ABSTRACT

While a growing number of technologies offer *personal data* to the user, little is known about how such tools can be harnessed for and by adolescents. Prior work has focused on implementations, in which youth experience is subordinated to the prescribed aims of adults. Few studies have engaged with the concerns and motives of the young people themselves. Co-design can be a powerful method for exploring beyond such limits, informing design that reflects the voices and values of the designed-for population. This paper presents a case study exploring the design of a personal informatics app (*LifeMosaic*) by a group of 14 to 16-year-olds. LifeMosaic lets users set any focus they personally care about and track it using colours and stickers. The design was shaped through youth concerns around privacy, social support, flexibility and self-expression as well as the desire to support mental wellbeing. This work illustrates new opportunities and understandings for personal informatics with youth: framing the data as mediating transindividual meaning making.

Author Keywords
personal informatics; co-design; youth-led design; meaning making.

CSS Concepts
• Human-centered computing ~ Empirical studies in interaction design.

INTRODUCTION

Personal informatics (PI) tools support users in collecting data about themselves for the purpose of later reflection [50]. PI can support the user's "self-understanding" of their own activity, mood, identity, relationships and other personally significant life factors [11; 12; 56; 68]. The commercial use of PI technologies (e.g. Fitbit, MyFitnessPal and RescueTime) is widespread, with many tools now native to smartphones and other devices. PI systems have been successfully adopted by different groups of adult users for chronic illness management [4; 51], patient-doctor discourse [14] and other motives [12; 19]; other work meanwhile has highlighted limitations of these systems, particularly where the goal is behaviour change [41; 54; 58]. The potential risks and benefits of PI for young people have long been recognised [20; 46; 65], but work here remains limited.

Prior work on uses of PI with youth broadly fall into two strands: framing the tools as either a teaching resource to illustrate STEM concepts [18; 47; 53; 69], or else as a part of a behaviour change intervention (for instance, nudging youth to be more active) [22; 25; 42; 81]. This focus on adult-defined outcomes has potentially restricted understanding of the perspectives and experiences of the young people themselves, as well as the exploration of other potential uses. Therefore, although adolescents are increasingly likely to interact with PI tools, a gap remains in the literature on how such experiences could be best harnessed and supported.

As Druin and others have noted [2; 27; 28], co-design can be a powerful method for voicing the concerns and understandings of young people. Allowing young people to "initiate and lead" [83] can challenge prior norms and assumptions and point to opportunities for future work. However, even within this approach, youth participants often indirectly inform the plans of a researcher, rather than acting as the "protagonists" of design [37]. The present study reports on a process of adolescent-led and initiated co-design of a PI app. This design team occupies a higher level on the "ladder of participation" [1] than is sometimes seen in IDC: the young people played an active role in the decision to do the work, the framing of what the work should focus on, and the direction of the overall project – enabling exploration and explication of uses, motives and responses that diverge from prior work in PI. The team's two-year dialectical process of collaborative meaning-making[1] and design is reported through images of the designed artefacts, interviews with team members and personal observations by the first author, as an adult member of the design team.

The central motivation for the present work was to find ways to empower young people in a process of design which they

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org.

IDC '20, June 21–24, 2020, London, United Kingdom
© 2020 Copyright is held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-1-4503-7981-6/20/06…$15.00
https://doi.org/10.1145/3392063.3394429

[1] i.e. motives and meanings emerged iteratively, in light of prior activity, whilst also shaping future activity [78].

find meaningful. The design activities and analyses were chosen to reflect the personal values and ideas of teens as experts on their own lives. The design team were asked to design a technology to make young people's lives better. As in prior work [30], the design team did not set out to create a personal informatics tool. Instead, personal informatics became a helpful framework for evaluating the design as their motives evolved.

This paper contributes to work facilitating youth-led co-design, illustrating the value of youth voices in guiding design choices. We suggest opportunities and challenges for future work bringing personal informatics to youth. We show this population can be keen and competent agents of PI practices: using their data in ways that are more expressive and polysemous than has been reported in prior work.

BACKGROUND

Personal Informatics

Developing, in part, out of self-care technologies that empower users to manage their health conditions at home [55; 74], personal informatics tools have tended to be adopted for self-improvement and behaviour change [41]. Though work in HCI has largely maintained this framing [52; 64; 79; 80], some strands have expanded the focus: noting that self-tracking is often a spontaneous and changing practice [66] which can support idiosyncratic motives, like dealing with the loss of a loved one [68] or communicating a sense of one's identity on a blind date [19]. These strands of work highlight how the meaning of personal informatics data is constructed by users, rather than given by the tool [7; 71].

Such user-led practices of meaning making have been supported through the design of PI tools which give more agency to the end user; for instance, through personalisation [39], flexibility [5] and semi-automated tracking [45]. This work has predominantly focused on adult users.

Personal Informatics with young people

Self-improvement

Whilst work with adult users of PI tools increasingly reports on and facilitates a discursive process of "self-understanding"[31], work with youth still tends to adopt a "self-improvement" framework. Most prior work in this context has implemented PI tools within behaviour change interventions, with youth participants borrowing a physical activity tracker and monitoring their own movement data to motivate progress towards a quantifiable outcome like increased physical activity or weight loss [22; 25; 42; 81]. Such approaches can leave little room for interpretation: the personal data here arguably functions as an extrinsic reward rather than necessarily supporting learning [40].

Some work within this strand has explicitly attempted to facilitate reflection and self-understanding. For instance, Garcia et al.[24] offered age-appropriate visualisations to students with ADHD to help them understand their distractions and develop resilience, by reflecting on how much they moved during the day's lessons. Here too however the behavioural outcome limited youth interpretations: it was predetermined that moving a lot was a problem and few details are reported on the youth users' own meaning making in response to these norms.

Curriculum learning

The potential of PI for learning has been explored more fully in a separate strand of work, implementing PI tools as a teaching resource: to teach a concept like *heart rate* or a skill like reading a graph [18; 47; 53; 69]. PI data visualisations have proven effective as personally-relevant stimuli to mediate classroom inquiry [18; 59]. Though beneficial to learning, such inquiry is still potentially limited by a prescribed outcome determined by the adult. The focus on fixed curriculum outcomes may relate to school priorities and constraints of the school day [49]. Chu et al. [13] aimed to involve youth participants in a freer process of inquiry by giving them smartwatches to take home. Though integration of the tools into everyday life was intended to support 'situated learning', this learning was evaluated by the extent to which tool use prompted thinking about science concepts – the authors report on the discordance between the school and home systems of meaning making for the PI practices. Prior work has noted the risk that such approaches restrict rather than expanding the scope of learning, reducing learner creativity [21].

There are also risks related to the potentially sensitive nature of PI practices with youth. In asking young people to collect personal data, issues have been raised about the risks of bullying or of the data being used by teachers or parents for uses not intended by the young person [20]. Some work has explored ways to address this gap; for instance, anonymizing youth data, through avatars [23] or giving the young person more control over what data is collected [61].

Agency and self-understanding

As Hong et al. [32] note, teens are still developing skills like communication and self-reflection. Their study suggests how shifting the focus of self-tracking from adult-defined to youth-defined outcomes can allow PI tools to scaffold self-expression. Whilst prior work has tended to focus on adult-defined curriculum content, some prior work suggests PI tools could help young people to learn about and express their own values, feelings and identities [30; 32; 48; 60]. As Lee et al. [48] note, this approach can also be valuable in better understanding the risks and challenges involved in bringing PI tools to youth. A key aspect of how this work has supported youth agency has been to involve young people in the design of the PI tool.

Figure 1: Setup screen. The user can choose a suggested focus to track or create their own.

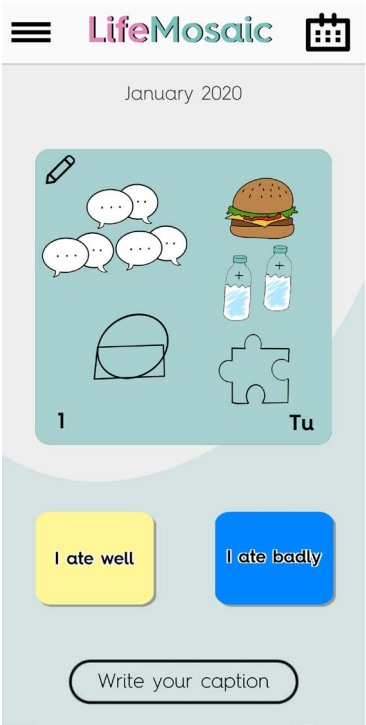

Figure 2: Daily tile screen. The user can log events with buttons which tint the tile, and add stickers for context.

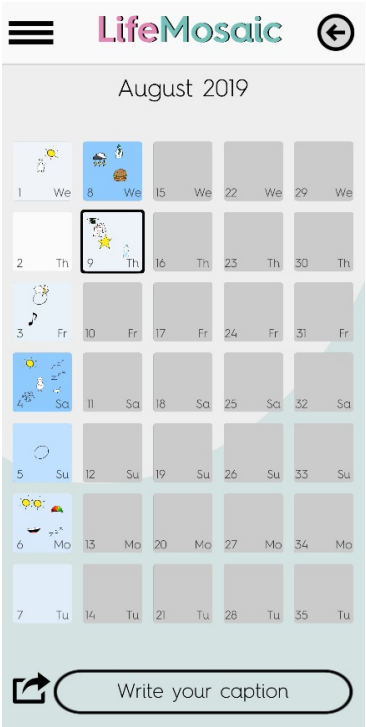

Figure 3: Mosaic screen. The user can see an overview of their month.

Co-design with young people

The value of giving young people agency in the design process has been widely reported in the IDC literature [26; 28; 37]. Co-design can give voice to the perspectives of young people as an often-ignored designed-for population [2; 16]. Co-design allows young people to explicate values and concerns which can lead further design, in a dialectical process [36]. However, the role of youth in the design process varies widely, with many co-design studies restricting design activities and the extent to which they directly shape design outcomes [37]. Youth in co-design of mobile technologies have often tended to be employed as reviewers (*informants*) rather than producers (*design partners*) of the designed artefacts [72]. Such restrictions could potentially affect the motivation of some youth participants, as well as the authenticity and depth of their contributions [8].When design partners are teenagers, it may be particularly important to enable them to pursue their own motives, collaborate meaningfully and develop a sense of ownership within the design process [6; 10; 35].

Co-design of personal informatics tools for teens has primarily focused on supporting self-care for a health condition [1; 32; 43; 63]. This work has typically employed young people as informants rather than design partners, but with notable exceptions [6; 30; 48]. Hodson et al. [30] report on the co-design of a tool to support adolescents experiencing a mental health crisis. The design work took place across four years, allowing youth design partners to play an active role at every stage of the process: directly contributing to the initial direction of the work as well as the final designed artefact. Ideation drew on the existing understandings and practices of the young people, including strategies for managing mental health crises. Youth creation of design elements was shaped by their own developing concerns and motives and the feedback of others in their community. This iterative process resulted in the creation of the *Mellow* app (a tool with PI elements: allowing the user to visualize and reflect on their mood). The co-design empowered a vulnerable population to make nuanced and meaningful contributions which reflected their own values. Potapov et al. [61] suggest that taking part in design work could develop youth understanding of PI practices, preparing youth to interpret their own PI data more creatively. Youth participants designed their own data visualisations before carrying out a month of self-tracking using commercially available PI tools. The authors report that youth participants gave creative interpretations of their own data – exploring personal interests, concerns and identities.

METHOD

In this study, we report on the design of a personal informatics tool by a group of 14 to 16-year-olds from varied socioeconomic backgrounds. The study focused on a design team of four UK secondary school students (3M, 1F). The small sample size limits the generalisations that can be drawn from the specific study context. We implemented a case-study design [17; 82] offering a qualitative account of the design process.

The design work began when participants were in Year 9 and continued through to Year 11, with one younger participant joining the team at the start of Year 10. To examine this long-term course of development, whilst foregrounding the motives, practices and perspectives of the youth participants, we adopt a Vygotskian framework [75; 77]. Designed artefacts are evaluated in the context of their role in mediating youth motives [33] i.e. we focused on how designed artefacts express youth concerns, rather than giving an adult evaluation of design affordances.

Data on the development of the design was collected "on two planes"[76]: first, through observations of the design work across 20 thirty-minute lunchtime club sessions; then in a twenty-minute semi-structured summative interview. Each team member was interviewed separately to foreground their own voice, explicating the concepts and concerns they had personally "internalized" [75; 76]. Interviews focused on design goals, rationale behind designed functions, and intended uses for the app. Sessions were audio-recorded and transcribed. Transcripts, observation notes and designed artefacts were thematically analysed [15]. Emerging themes were member checked and modified based on feedback [29].

We quote extensively from the young people in characterising the design process. This is motivated by our study aims of privileging youth voices. The summative interviews help frame the observational data from the young people's perspectives.

Process of co-design

Participants and initial motivations

The design work in this study was framed by the team's involvement in Nesta's Longitude Explorer Prize. The prize encourages "youth-led innovation" in STEM to address real-world problems [67]. The 2017 prize invited young people across the UK to design an Internet of Things product to make people's lives better. Responding to a notice about the chance to enter a STEM competition, students attended a specially formed weekly lunchtime club to work on their prize entry. A dozen or so students attended regularly until the submission. These students could be said to be more academic and self-determined than their peers.

Before considering design or existing technologies, the group discussed problems they know people around them face in their everyday lives. A number of problem areas like sedentary lifestyle and junk food were suggested but the group quickly came to a consensus around the issue of teen mental health. *"It was just something that came out of organic discussion about what would be best for someone in a situation similar to us."* (Seth). Several club attendees shared experiences of helping friends going through mental health issues. They felt such issues were widespread and that there was a *"health crisis, both physically and mentally, in young people and teenagers and a lack of resources given by public services [to] deal with crisis situations."* (Xian)

After researching existing IoT tools, they came up with *"a smartwatch that could support mental health… [by launching] your personal coping strategies."* (Geordie). The students suggested that adolescents with anxiety may have a specific person, video or other technique to help manage negative mental health symptoms. The conceived smartwatch would launch this strategy during a panic attack. This idea was submitted for the prize but did not advance to the next round. However, three members of the group were keen to continue working on the idea. Together with a fourth student, who joined six months later, they formed the design team of the present study. Other students contributed to the design work at different stages but chose not to be involved at the time of the interviews.

As the team worked on the interface of their wearable device, their research expanded to existing mood and mental health apps. The first author introduced the concept of *self-tracking* as a useful way to characterize the kinds of features the team were exploring. Discussions of these tools and of their own experiences using apps like Samsung Health and Daylio informed a change in design focus: from a wearable to a personal informatics app *"where you set the variables and kind of get into it without having to fork out a lot of money or do anything special."* (Geordie). As well as shaping these concerns about ease of use and accessibility, the discussions opened up the focus of ideation from mental health treatments and other medical themes to a more casual and holistic view of wellbeing and flourishing, and the factors that could impact them. This initial work framed the design of the LifeMosaic PI tool.

Co-design of a personal informatics tool

The work of the design team continued in weekly lunchtime club sessions. Though the framing in the preceding sub-section should be noted, a deliberate effort was made by the first author, as club facilitator, not to direct the course of design. Design choices were based on the young people's own lived experiences.

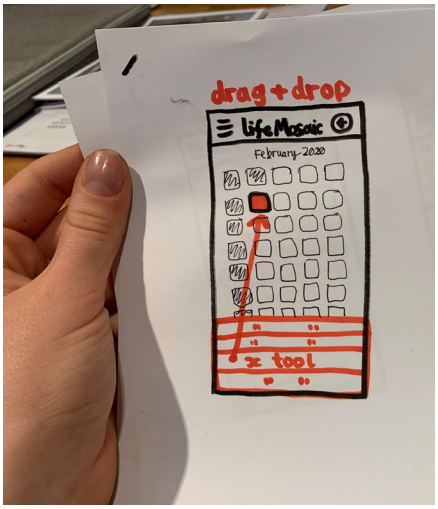

Figure 4: Design idea for mosaic screen.

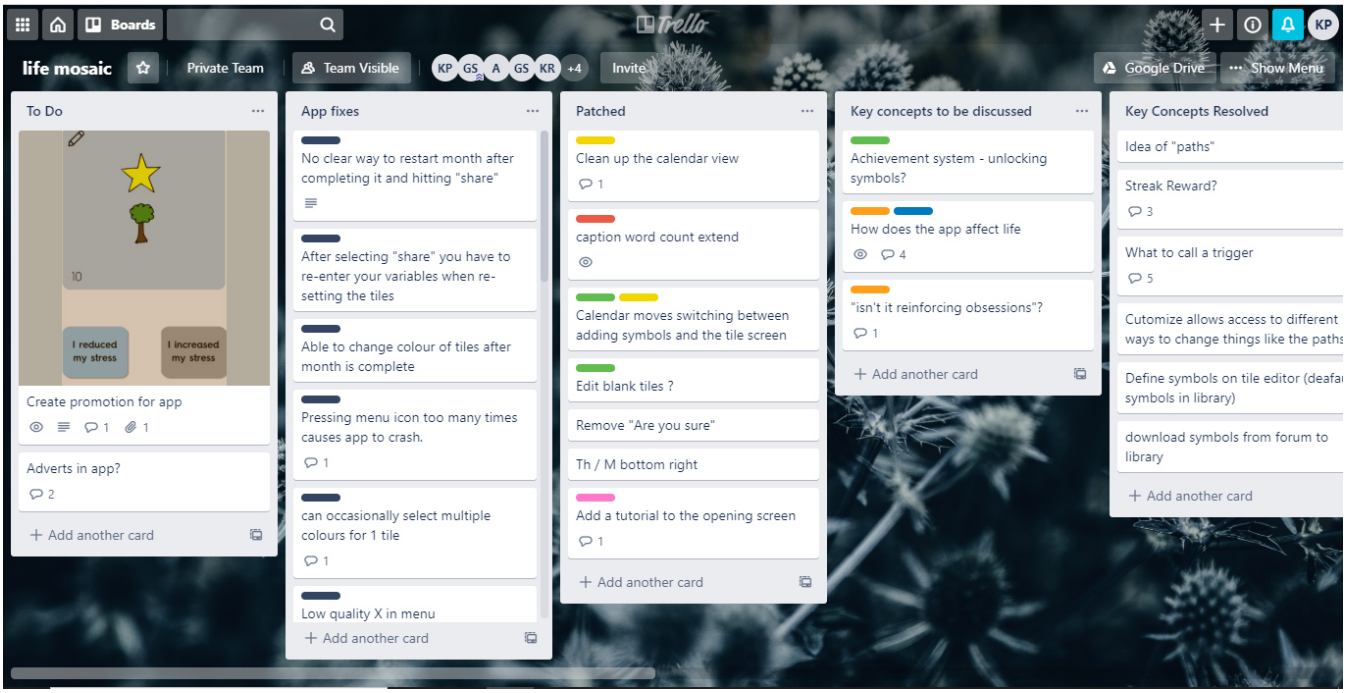

Figure 5: Trello board of tasks and ideas for the app.

Lunchtime club sessions involved discussions around specific problems and app features. The freeform structure meant that some sessions dealt with abstract concepts like *online bullying* or *progress*, while others dwelt on the position of a button or what to call a menu item. The team often created design fictions for "future versions" or spent weeks on a design feature they later deemed unrealistic or unnecessary. As the work progressed, designed artefacts created between sessions (Figure 4) or drawn on mini whiteboards during the session helped mediate the discussion: clarifying ideas and focusing negotiations in a disagreement.

As the design grew more complicated, the club facilitator showed the team how to set up a Trello board (Figure 5). This greatly helped the team to structure their design work and added a layer of accountability which did not require direct intervention by the facilitator. The team assigned tasks to each-other using this board and completed these tasks at home between sessions.

The club facilitator's principal role in the design team was to facilitate discussion: keeping focus and highlighting emerging questions. Here, care was taken to seek clarification or prompt with ideas already in the youth discourse, rather than dismissing contributions or intervening with new content [36]. Once Trello had been introduced, the facilitator acted as "minute keeper", helping the team put their tasks and ideas onto Trello cards (though team members often rearranged these and added their own between sessions). Having the team Trello on the interactive whiteboard allowed weekly follow-ups of the assigned tasks, and structured subsequent discussions. Problems and disagreements could be quoted on a Trello card to be returned to later.

The facilitator also managed external links and took on roles that can only be carried out by an adult: setting up KickStarter and Upwork accounts; finding and liaising with professionals for specific consultation; recruiting students for additional design work and testing.

As the design developed, local parents offered their support and gave the team workshops in fundraising, branding and app specification writing. The team made use of this valuable advice to publicize the app and take it to a developer. Though the team sought advice and guidance at different stages of the design processes, the designed elements submitted to the app developer were wholly designed by the young people themselves. LifeMosaic was built when the team submitted their designed elements, specification and user stories to a freelance developer, following a successful crowdfunding campaign supported by the local community.

THE LIFEMOSAIC APP

LifeMosaic is a personal informatics smartphone app that lets the user focus on any aspect of their life and log it daily using colours and symbols (*stickers*). The user tracks two opposing actions or qualities, such as "I slept well: I slept badly", "I made my mood better: I made my mood worse" or "I revised: I went on YouTube" (Figure 1). These are represented on two coloured buttons. The user clicks the button to indicate how many times or to what extent an event has happened. Pressing combinations of the two buttons blends their assigned colours on a *daily tile*. (Figure 2) The shade of the daily tile indicates to the user where they are along the spectrum of their chosen focus. The user can add

stickers to their daily tile from the sticker library. These symbolize contexts like what triggered an event, what activities the user has done, or how they were feeling. The user can also create their own simple sticker on a sticker design screen.

The user can view a visualisation of their month (their *mosaic*), composed of their daily tiles and aligned vertically week to week (Figure 3). An image of the *mosaic* can be archived or shared with others.

KEY MOTIVES AND DESIGN CHOICES

The primary design goal of supporting teenagers in learning about their own wellbeing permeates the team's discussion of specific app functions. These design choices have been coded into four sub-motives: enabling customization, supporting data interpretations, facilitating conversations and maximizing privacy. The design team agree that LifeMosaic captures their intentions and negotiations over the course of the two years. The design crystallizes their practices [34; 35] and the motives, values and contradictions behind them [30; 36].

Enabling customization

It was key throughout the design work that as a user you should be able to "make the app your own" (Geordie). The team felt that the lives of young people were too varied and fast-changing for the "rigid" design of existing PI tools.

So, for some people doing an hour's exercise might be ridiculous. So you can say that you've done 10 minutes and still be proud of yourself and still be able to put positive input in and see for yourself that that's something you've done well… or another week just not shutting yourself off in your room is worth recording. (Geordie)

The PI tool needed to be flexible enough to record potentially subtle and changing factors valued in everyday life activity of the young user.

[Other PI tools] defined for you what happy and sad was as opposed to you can say yourself what the behaviours are that are making you happy or sad… or if that's not how you see it: if that's not for you then you can do 'social/antisocial', 'helpful/ unhelpful' for instance. (Seth)

The team felt it was important for the user to be able to set not only the personal data type they were tracking but also the criteria by which they evaluated that data. In contrast to behaviour change framings of PI systems (in which it is a given, for instance, that a higher step count is of value) the team wanted their PI tool to allow them to define the framing of interpretation as well as the data source: what was qualified as well as what was quantified. Applying a Vygotskian lens, the intention here was seemingly to allow for the data to align with existing ideals/systems of judgement that affected the user, allowing them to develop [77]: *"You can find things that work for you and people who help you but you need to do the work to understand how to make progress."* (Seth). In contrast to adult users, who often acquire PI tools to make progress towards known instrumental ends [41], Seth wanted his PI tool to be part of wider process of learning about what constitutes progress for him.

On first launching LifeMosaic the user sees a setup screen (Figure 1) on which they choose a "tracking focus" and what it is about that focus that they want to evaluate. This is represented through two contrasting actions or judgements. Suggested options include "I was social: I was not social" and "My day went well: My day went badly". The user can also easily create their own tracking focus. This allow the user to set their own system for interpreting life factors. For example, the user could focus on mood and evaluate this as "My day went well", "I had good times" or "I feel positive" or a subfactor like "I felt calm". The user can use sticker symbols (Figure 2) to separate what they take as data and what as judgement. For example, they could use the symbols to represent people they spoke to or activities they've done, and then use the "I was social: I was not social" buttons to judge/interpret this data.

Self-determination and empowerment were key values explored in the design work. The design was shaped by their motive to empower the user *"to figure out how they can take control of their own life"* (Geordie). The team felt that giving the user agency in their meaning making could support wellbeing and self-understanding.

A lot of the time we found existing apps were extremely focused and narrow upon some things such as mental health or losing weight, whereas LifeMosaic aims to put the user in control - expressing themselves in a way that's more meaningful to them… so there isn't that kind of pressure, which for teenagers is actually probably what you're trying to avoid. (Xian)

Rather than saying this is how you should change or even this is bad or this is good, the most important thing was how can you understand for yourself and just learn more about yourself every day. (Seth)

The team acknowledged the value of systems and routines for personal growth, but did not want their PI tool to tell them what to do: in part because this would not capture their individual values and in part because it disempowered them along their long and tentative project of self-understanding. For a teen population who are still developing a sense of identity, it was important that the app could support them in learning about themselves whilst giving them room to experiment.

Supporting interpretations

The team wanted to prioritize ease of use and understanding for the user. An early challenge concerned the visualization of the data, with trade-offs discussed between simplifying what the user saw and capturing what was important. As Geordie noted, apps like RescueTime *"can show you a load of stuff [that took] too much effort"* to access and understand. There was a worry that when users are teens, seeing "a graph

might not feel that relevant" (Esme), prompting abandonment before the user has invested time interpreting potential significance.

Together with the concerns around customization, these motives directed design towards visualizations that were playful [62] rather than technical or scientific and which could be interpreted in multiple ways. *"Because [when] it's all about numbers and is more precise data, it kind of restricts what the user can do on the app. And it gives you one answer whereas on LifeMosaic you can have many different answers."* (Esme)

Data collection on LifeMosaic centred around two buttons, corresponding to the two poles of the tracking focus e.g. "I slept badly: I slept well". The user assigned a colour to each pole and their daily tile visualization was tinted with that colour on each button press (Figure 2). This created a visualization which was expressive rather than precise: the user would not know how many times they pressed each button but the resulting colour would give them a sense of where they were on the gradient between the two poles.

This work helped explicate the value of self-expression and creativity for the design. The team wanted the user to express as much as possible about what was significant for them. The team designed a library of stickers that could be dragged onto the visualization to signify whatever the user wanted it to. These symbols were designed to be whimsical and open to taking on different meanings. This gave the user the flexibility of *"Not having to say, you know, this sticker means I'm happy, but the tree could mean that erm you fell down the stairs. And being able to see, you know if there's quite a lot of trees, you're quite clumsy."* (Geordie) For Geordie, the sticker did not have a fixed representational value; its value was produced through its ideality [33] i.e. its relation to everyday life activity.

LifeMosaic offers a monthly visualizations of the user-created tiles (Figure 3) which let the user look "back on your day, your month or your week and kind of see what affects you… how things are going" and potentially notice "trends" (Seth) like weeks or days of the week where similar significances were expressed. The process of meaning making was potentially supported here through the fact that the user had already invested meaning in choosing how to represent the data. This addressed the barrier of interpretation through the close correspondence between data collection and meaning making: the data was interpreted to express personal significance in its collection.

Over time, the user could create their own system, using the sticker symbols to convey varying levels of abstraction.

"This tile I've drawn a burger with an arrow facing up and a bottle with an arrow facing down because I ate more fast food today and drank less and I put a spanner at the bottom to remind myself it's something I want to work on and I want to fix it." (Xian)

The personalized creativity here could potentially motivate youth users put off by the onus of interpreting more traditional graphs and visualizations.

Facilitating conversations

Much of the play and creativity in the interpretation of PI data visualizations emerged in the design work through the dynamics of discourse and collaboration. After some time experimenting with PI tools, the team wanted to find ways to reproduce such social dynamics for the user. The values driving these concerns related chiefly not to sociability but to support and empowerment. The expressive nature of the data reflected the team's concern for the visualizations to be something that could be shared and talked about (with a friend or mentor). *"So how I can tell Seth about my week on LifeMosaic you can have someone you know you can talk to."* (Geordie) The team hoped this could help the student feel less *"pressure [than seeing a] psychiatrist or a therapist or someone they aren't entirely comfortable sharing it with."* (Geordie). The data was seen as a ticket for meaningful discourse that could support inquiry and self-understanding.

Instagram for instance believes that social desirability and having many friends is the most valuable thing and they use a likes system to represent that, whereas LifeMosaic is purely based around recording and opening discussions… in the end we just aim to facilitate people connecting with other people about who they are. (Xian)

The team had had positive and enlightening experiences on social media and recognized its power but were critical of the token economies of "likes", which they felt got in the way of genuine and meaningful connections. Through this lens, they came to see a PI visualization as a potentially more authentic social media post. At the time of the interviews work had begun on a private forum where users could anonymously post their 'mosaic' with a narrative about it. Other users would click "relate" to indicate they felt a connection with something in the post.

It would help you understand that it is not just happening to you and it's happening to lots of other people. So you wouldn't feel as targeted or victimized by powers out of your control. (Esme)

The team recognized a tension between young people's fear of being judged or bullied, and the value of the support of their peers, and tried to mediate this in design.

Maximising privacy

The concern for social sharing was underscored by an equal appreciation of the need to protect users from the risks of their personal data being misused and *"certain stuff being seen by a teacher"* (Geordie) or other unwanted audiences.

So other apps are starting to like become corrupt and their sharing features make it public so your information is a lot less personal and you don't know where it's going… [someone] can steal it or pass it on. (Esme)

It was important for the user of LifeMosaic to have full control over who saw their data. The "share" function (Figure 3) took a screenshot of the mosaic but this visualization did not offer any identifying or personal information. The use of abstract symbols could allow the user to develop their own codes, stopping the data from being meaningful to anyone they did not choose to discuss it with.

DISCUSSION

Personal informatics tools have tended to be designed for a narrow demographic of adult users [70], with teen populations rarely considered [9]. Prior work bringing PI to youth has tended to focus on particular short-term outcomes related to behaviour change, health management or knowledge transfer [23]. Reporting on youth-led design that did not adopt such framing, this paper describes teen uses of PI tools that divert from these norms; these include social connection, identification of personal obstacles and self-expression. These findings contribute to work on PI with youth, suggesting opportunities for PI to support personally significant motives.

This paper highlights how the motives and responses of youth users may differ from those of adult users. While youth motives are shaped through the institutions in which they participate, they also extend beyond them [35]. Prior work has often ignored these motives by restricting youth activity: the contributions of youth users have tended to be evaluated against a curriculum focus or behavioural outcome. Our findings indicate that teens may benefit from being given a more active role in the development of such criteria. The LifeMosaic team wanted to set their own standards for judging their personal data: evaluating whether they were productive, sociable or another quality they felt significant in their everyday lives.

Rather than seeing PI as a behaviour change intervention, the LifeMosaic design team considered how their app could be integrated into their existing practices as part of a tentative and ongoing process of self-understanding. They did not want their PI tool to tell them what to do (Seth) or act as an extension of the institutional and social pressures they already faced (Xian). Above all, they valued agency and self-determination and wanted their PI tool to support their own meaning making in spheres of experience beyond the classroom [57].

While early work in PI has proposed that self-tracking practices could support student reflection on personal values [60], this area remains underexplored. Our findings address this gap, suggesting that PI tools can help youth to express and shape personal values in social contexts. Flexibility and self-expression was valued above accuracy and technology-generated interpretations. LifeMosaic was focused on *"directly communicating someone's experience of the world"* (Xian). The team felt LifeMosaic could empower the user by helping them learn about and express aspects of their fast-changing lives, values and identities. They felt existing tools were too rigid to allow this.

In contrast to traditional personal informatics tools, LifeMosaic gives the user an active role in designing their data and constructing its meaning. Whilst prior work has recognised the desire for such flexibility [5; 44] among users, existing PI tools focused on user agency and customization [45] are often too complex, especially for young users. Some designers have responded to this limitation by creating PI data visualisations that are expressive and casual [62] rather than precise and rigorous. Storni's *Tag-it-Yourself* app prototype was developed to empower adults with diabetes to record their idiosyncratic and fast-changing concerns and practices, letting users create custom and multimedia tags to log everyday life activities [73]. Similarly, Ayobi et al.'s *Trackly* harnesses the practices of bullet journaling and colouring to give users with MS the agency to easily choose what they want to track and how to visualize it [3]. LifeMosaic contributes to these efforts, illustrating opportunities for future work to better scaffold customisation without limiting the user's creativity and sense of agency. Adopting a cultural-historical lens, one could say that when a user can modify an object like a data visualisation through their own actions, this object can come to embody a complex system of values and practices [34; 35; 78].

These findings contribute to work on PI with youth examining how personal data visualisations can act as an 'object of inquiry' to support student-led learning and creativity [13; 59]. Rather than seeing the meaning of the data as given by the tool, the design team portrayed it as open to multiple interpretations, through their own (dialectically developing [38]) judgements. The collaborative process of co-design allowed the data to become rich and evocative. The data could take on values and motives unique to the user's activity [78]. The LifeMosaic user could record *"an art pallet to show I want to create more relationships"* (Xian) and use this act as part of personal exploration, to ask for support or to connect with others. These findings illustrate a potential limitation in prior work in which the meaning of a sign in a PI practice is fixed by an adult before it can be appropriated for a problem by the young person [75].

Following prior work with adults [19], our findings demonstrate that when personal data is framed as playful and expressive, it can act as a 'ticket to talk', helping the user to connect with others and share aspects of their identity. Though schools are inherently social places, prior work with PI in schools has tended to follow the conventions of PI with adult users [50], treating it as an individual rational activity [23]. Our findings address this gap and motivate future work: showing that peer-to-peer interactions could be a powerful tool for supporting and enriching youth meaning making.

The gap in prior work considering youth motives of PI users in schools may relate to the challenge of managing the wide range of voices in a classroom, given the constraints of the school day. Differentiating for the concerns and motives of youth participants may present a barrier for research [53].

Our findings indicate that young people may value such flexibility and collaborate to support each other's activities and learning: a greater degree of agency over the focus of the youth practice may address the limitation.

Whilst the design team valued the support of peers, they recognised the dangers of sharing personal and potentially sensitive details with others. Their design choices prioritized privacy and curation; ideas like that of a private forum attempted to negotiate the tension between supporting social connections and protecting young users from harm. Contributing to prior work [23], LifeMosaic explores how youth identities can be anonymised while allowing the data to be shared as a public object of inquiry. The use of ambiguous symbols and personalised systems of evaluation could suggest some further directions for design addressing this important challenge. Future work should consider how youth self-tracked personal data can be curated by the young person and protected from misuse, without restricting its social value.

Design has often ignored the perspectives of end users and this is particularly true in design for young people. Though co-design has proven a powerful method for explicating youth values [36], work here can be limited in the agency afforded to participants [8]. The present paper contributes to work in this area by reporting on design with adolescents whose contribution is higher on the "ladder of participation" [1] than is typical in co-design with youth. Our findings demonstrate the value of this approach for gaining insight into youth concerns and revealing new opportunities for research. Following prior work [16], we show long-term co-design can empower youth to express personal concerns, motives and identities; breaking away from standard norms and curricula, youth are able to learn about themselves.

SELECTION AND PARTICIPATION OF CHILDREN

Since this paper reports retrospectively on work which young people began of their own volition, care was taken to separate their continued participation in the design work as part of their school activity and their participation in the study. Interviews were conducted one on one outside of school hours to ensure no team member felt pressured to participate. Parents of participants completed consent forms and participants completed forms of assent, approved by an institutional ethics board. Emerging themes from the interviews were discussed with the team and a copy of this paper shared with them and their parents. Under the participants' request and with parents' consent, the real names of team members have been used.

ACKNOWLEDGEMENTS

First, of course, we would like to thank Xian, Esme, Geordie and Seth as our collaborators, as well as to the other young people who helped create this unique technology. Second, thank you to Matt, Jules, David and the other adults who gave up their time to give the team workshops and advice. Thank you also to the school, and to the parents and local people who helped raise funds to build the app, with special thanks to Sally Field and Barclays Eagle Labs for their generous donations. A huge thank you to Raj who took the designs and patiently listened to what the young people wanted in order to build the app. Finally, thank you to the anonymous reviewers for their valuable feedback on an earlier draft of this paper.

REFERENCES

[1] Lars Bo Andersen, Peter Danholt, Kim Halskov, Nicolai Brodersen Hansen, and Peter Lauritsen, 2015. Participation as a matter of concern in participatory design. *CoDesign 11*, 3-4, 250-261.

[2] Theresa Anderson, Allison Druin, Kenneth Fleischmann, Eric Meyers, Lisa Nathan, and Kristene Unsworth, 2009. Children, technology and social values: Enabling children's voices in a pluralistic world. *Proceedings of the American Society for Information Science and Technology 46*, 1, 1-9.

[3] Amid Ayobi, Paul Marshall, and Anna Cox, 2020. Trackly: A Customisable and Pictorial Self-Tracking App to Support Agency in Multiple Sclerosis Self-Care. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI'20)* ACM.

[4] Amid Ayobi, Paul Marshall, Anna L Cox, and Yunan Chen, 2017. Quantifying the Body and Caring for the Mind: Self-Tracking in Multiple Sclerosis. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* ACM, 6889-6901.

[5] Amid Ayobi, Tobias Sonne, Paul Marshall, and Anna L Cox, 2018. Flexible and Mindful Self-Tracking: Design Implications from Paper Bullet Journals. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* ACM, 28.

[6] Adam Bell and Katie Davis, 2016. Learning through participatory design: designing digital badges for and with teens. In *Proceedings of the The 15th International Conference on Interaction Design and Children* ACM, 218-229.

[7] Matthias Bode and Dorthe Brogård Kristensen, 2016. The digital doppelgänger within. A study on self-tracking and the quantified self movement. *Assembling consumption: Researching actors, networks and markets*, 119-134.

[8] Tone Bratteteig and Ina Wagner, 2016. Unpacking the notion of participation in participatory design. *Computer Supported Cooperative Work (CSCW) 25*, 6, 425-475.

[9] Carme Carrion, Maurizio Caon, Stefano Carrino, Liliana Arroyo Moliner, Alexandra Lang, Sarah Atkinson, Marco Mazzola, Paolo Perego, Carlo Emilio Standoli, and Conxa Castell, 2015. Wearable lifestyle tracking devices: are they useful

for teenagers? In *Adjunct Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2015 ACM International Symposium on Wearable Computers* ACM, 669-674.

[10] Bester Chimbo and JH Gelderblom, 2014. Comparing young children and teenagers as partners in co-design of an educational technology solution. In *Proceedings of the ISI e-Skills for Knowledge Production and Innovation Conference*, 17-21.

[11] Eun Kyoung Choe, Bongshin Lee, Haining Zhu, Nathalie Henry Riche, and Dominikus Baur, 2017. Understanding self-reflection: how people reflect on personal data through visual data exploration. In *Proceedings of the 11th EAI International Conference on Pervasive Computing Technologies for Healthcare* ACM, 173-182.

[12] Eun Kyoung Choe, Nicole B Lee, Bongshin Lee, Wanda Pratt, and Julie A Kientz, 2014. Understanding quantified-selfers' practices in collecting and exploring personal data. In *Proceedings of the 32nd annual ACM conference on Human factors in computing systems* ACM, 1143-1152.

[13] Sharon Lynn Chu, Brittany Garcia, and Beth Nam, 2019. Understanding Context in Children's Use of Smartwatches for Everyday Science Reflections. In *Proceedings of the 18th ACM International Conference on Interaction Design and Children*, 83-93.

[14] Chia-Fang Chung, Kristin Dew, Allison Cole, Jasmine Zia, James Fogarty, Julie A. Kientz, and Sean A. Munson, 2016. Boundary Negotiating Artifacts in Personal Informatics: Patient-Provider Collaboration with Patient-Generated Data. In *Proceedings of the Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing* (San Francisco, California, USA2016), ACM, 2819926, 770-786. DOI= http://dx.doi.org/10.1145/2818048.2819926.

[15] Victoria Clarke and Virginia Braun, 2014. Thematic analysis. In *Encyclopedia of critical psychology* Springer, 1947-1952.

[16] Merijke Coenraad, Jen Palmer, Diana Franklin, and David Weintrop, 2019. Enacting Identities: Participatory Design as a Context for Youth to Reflect, Project, and Apply their Emerging Identities. In *Proceedings of the 18th ACM International Conference on Interaction Design and Children*, 185-196.

[17] John W Creswell and Dana L Miller, 2000. Determining validity in qualitative inquiry. *Theory into practice 39*, 3, 124-130.

[18] Joel R Drake, Ryan Cain, and Victor R Lee, 2018. From Wearing to Wondering: Treating Wearable Activity Trackers as Objects of Inquiry. In *Wearable Technologies: Concepts, Methodologies, Tools, and Applications* IGI Global, 810-832.

[19] Chris Elsden, Bettina Nissen, Andrew Garbett, David Chatting, David Kirk, and John Vines, 2016. Metadating: exploring the romance and future of personal data. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* ACM, 685-698.

[20] Bård Ketil Engen, 2016. Wearable computer: what's in it for schools? Opportunities and risks.

[21] Rebecca Eynon, 2015. The quantified self for learning: critical questions for education Taylor & Francis.

[22] Luis Fernandez-Luque, Meghna Singh, Ferda Ofli, Yelena A. Mejova, Ingmar Weber, Michael Aupetit, Sahar Karim Jreige, Ahmed Elmagarmid, Jaideep Srivastava, and Mohamed Ahmedna, 2017. Implementing 360° Quantified Self for childhood obesity: feasibility study and experiences from a weight loss camp in Qatar. *BMC Medical Informatics and Decision Making 17*, 1 (April 13), 37. DOI= http://dx.doi.org/10.1186/s12911-017-0432-6.

[23] Andrew Garbett, David Chatting, Gerard Wilkinson, Clement Lee, and Ahmed Kharrufa, 2018. *ThinkActive: Designing for Pseudonymous Activity Tracking in the Classroom*.

[24] Juan Jimenez Garcia, Hilde de Bruyckere, David V. Keyson, and Natalia Romero, 2013. Designing Personal Informatics for Self-reflection and Self-awareness: The Case of Children with Attention Deficit Hyperactivity Disorder Springer International Publishing, Cham, 109-123.

[25] Jeffrey Gaudet, François Gallant, and Mathieu Bélanger, 2017. A Bit of Fit: Minimalist Intervention in Adolescents Based on a Physical Activity Tracker. *JMIR Mhealth Uhealth 5*, 7 (2017/07/06), e92. DOI= http://dx.doi.org/10.2196/mhealth.7647.

[26] Lalya Gaye and Atau Tanaka, 2011. *Beyond participation: empowerment, control and ownership in youth-led collaborative design*. Association for Computing Machinery, Atlanta, Georgia, USA.

[27] Helene Gelderblom, 2014. Giving children voice in the design of technology for education in the developing world. *South African Computer Journal 54*, Special issue 2, 34-42.

[28] Mona Leigh Guha, Allison Druin, and Jerry Alan Fails, 2013. Cooperative Inquiry revisited: Reflections of the past and guidelines for the future of intergenerational co-design. *International Journal of Child-Computer Interaction 1*, 1, 14-23.

[29] Lou Harvey, 2015. Beyond member-checking: A dialogic approach to the research interview. *International Journal of Research & Method in Education 38*, 1, 23-38.

[30] Elise Hodson, Nastaran Dadashi, Ramon Delgado, Connie Chisholm, Robert Sgrignoli, and Ronald Swaine, 2019. Co-design in mental health; Mellow: a self-help holistic crisis planning mobile application by youth, for youth. *The Design Journal 22*, sup1, 1529-1542.

[31] Victoria Hollis, 2018. On Being 'Fitter, Happier, and More Productive': The Impact of Implicit Goals in Affective Personal Informatics UC Santa Cruz.

[32] Matthew K Hong, Udaya Lakshmi, Thomas A Olson, and Lauren Wilcox, 2018. Visual ODLs: Co-designing patient-generated observations of daily living to support data-driven conversations in pediatric care. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1-13.

[33] Ėvald Vasilevich Ilyenkov, 1974. Activity and knowledge. *Philosophy and culture*.

[34] Ėvald Vasilevich Ilyenkov, 2009. *The ideal in human activity*. Marxists Internet Archive Pacifica (CA).

[35] Ole Sejer Iversen, Christian Dindler, and Elin Irene Krogh Hansen, 2013. Understanding teenagers' motivation in participatory design. *International Journal of Child-Computer Interaction 1*, 3 (2013/09/01/), 82-87. DOI= http://dx.doi.org/https://doi.org/10.1016/j.ijcci.2014.02.002.

[36] Ole Sejer Iversen, Kim Halskov, and Tuck W Leong, 2012. Values-led participatory design. *CoDesign 8*, 2-3, 87-103.

[37] Ole Sejer Iversen, Rachel Charlotte Smith, and Christian Dindler, 2017. Child as protagonist: Expanding the role of children in participatory design. In *Proceedings of the 2017 Conference on Interaction Design and Children* ACM, 27-37.

[38] Vera John-Steiner and Holbrook Mahn, 1996. Sociocultural approaches to learning and development: A Vygotskian framework. *Educational psychologist 31*, 3-4, 191-206.

[39] Ravi Karkar, Jasmine Zia, Roger Vilardaga, Sonali R Mishra, James Fogarty, Sean A Munson, and Julie A Kientz, 2016. A framework for self-experimentation in personalized health. *Journal of the American Medical Informatics Association 23*, 3, 440-448.

[40] Charlotte Kerner and Victoria A. Goodyear, 2017. The Motivational Impact of Wearable Healthy Lifestyle Technologies: A Self-determination Perspective on Fitbits With Adolescents. *American Journal of Health Education 48*, 5 (2017/09/03), 287-297. DOI= http://dx.doi.org/10.1080/19325037.2017.1343161.

[41] Elisabeth T. Kersten-van Dijk, Joyce H. D. M. Westerink, Femke Beute, and Wijnand A. Ijsselsteijn, 2017. Personal Informatics, Self-Insight, and Behavior Change: A Critical Review of Current Literature. *Human–Computer Interaction 32*, 5-6 (2017/11/02), 268-296. DOI= http://dx.doi.org/10.1080/07370024.2016.1276456.

[42] Eeva Kettunen, Markus Makkonen, Tuomas Kari, and Will Critchley, 2019. Using Sport and Wellness Technology to Promote Physical Activity: An Intervention Study among Teenagers. In *Proceedings of the 52nd Hawaii International Conference on System Sciences*.

[43] Katherine K Kim, Holly C Logan, Edmund Young, and Christina M Sabee, 2015. Youth-centered design and usage results of the iN Touch mobile self-management program for overweight/obesity. *Personal and Ubiquitous Computing 19*, 1, 59-68.

[44] Nam Wook Kim, Hyejin Im, Nathalie Henry Riche, Alicia Wang, Krzysztof Gajos, and Hanspeter Pfister, 2019. DataSelfie: Empowering People to Design Personalized Visuals to Represent Their Data. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* ACM, 79.

[45] Young-Ho Kim, Jae Ho Jeon, Bongshin Lee, Eun Kyoung Choe, and Jinwook Seo, 2017. OmniTrack: A flexible self-tracking approach leveraging semi-automated tracking. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies 1*, 3, 1-28.

[46] Victor R Lee, 2013. The Quantified Self (QS) movement and some emerging opportunities for the educational technology field. *Educational Technology*, November-December 2013, 39.

[47] Victor R Lee, 2019. On researching activity tracking to support learning: a retrospective. *Information and Learning Sciences 120*, 1/2, 133-154.

[48] Victor R Lee and Mary Briggs, 2014. Lessons learned from an initial effort to bring a quantified self" meetup" experience to a new demographic. In *Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing: Adjunct Publication*, 707-710.

[49] Victor R Lee, Joel R Drake, Ryan Cain, and Jeffrey Thayne, 2015. Opportunistic uses of the traditional school day through student examination of Fitbit activity tracker data. In *Proceedings of the 14th International Conference on Interaction Design and Children* ACM, 209-218.

[50] Ian Li, Anind Dey, and Jodi Forlizzi, 2010. A stage-based model of personal informatics systems. In *Proceedings of the SIGCHI conference on human factors in computing systems* ACM, 557-566.

[51] Haley MacLeod, Anthony Tang, and Sheelagh Carpendale, 2013. Personal informatics in chronic illness management. In *Proceedings of Graphics Interface 2013* Canadian Information Processing Society, 149-156.

[52] Mayank Malhotra, 2015. An investigation of usage barriers for personal informatics tools.

[53] Ivana Matteucci, 2017. Information About Ourselves from Ourselves: Young Users of Wearable Technologies in Secondary School. *European Journal Of Interdisciplinary Studies 9*, 1, 42.

[54] Sean A Munson, 2017. Rethinking assumptions in the design of health and wellness tracking tools. *interactions 25*, 1, 62-65.

[55] Francisco Nunes, Nervo Verdezoto, Geraldine Fitzpatrick, Morten Kyng, Erik Grönvall, and Cristiano Storni, 2015. Self-care technologies in HCI: Trends, tensions, and opportunities. *ACM Transactions on Computer-Human Interaction (TOCHI) 22*, 6, 33.

[56] Mika Pantzar and Minna Ruckenstein, 2015. The heart of everyday analytics: emotional, material and practical extensions in self-tracking market. *Consumption Markets & Culture 18*, 1, 92-109.

[57] Dorian Peters, Rafael Calvo, and Richard Ryan, 2018. Designing for Motivation, Engagement and Wellbeing in Digital Experience. *Front Psychol 9*(05/28). DOI= http://dx.doi.org/10.3389/fpsyg.2018.00797.

[58] Lukasz Piwek, David A Ellis, Sally Andrews, and Adam Joinson, 2016. The rise of consumer health wearables: promises and barriers. *PLoS medicine 13*, 2.

[59] Joseph L Polman, 2018. From quantified self to building a more fit community: data tracking and science infographics as boundary objects International Society of the Learning Sciences, Inc.[ISLS].

[60] Alina Pommeranz, Christian Detweiler, Pascal Wiggers, and Catholijn M Jonker, 2011. Self-reflection on personal values to support value-sensitive design. In *Proceedings of the 25th BCS Conference on Human-Computer Interaction* British Computer Society, 491-496.

[61] Kyrill Potapov, Victor R Lee, Asimina Vasalou, and Paul Marshall, 2019. Youth Concerns and Responses to Self-Tracking Tools and Personal Informatics Systems. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* ACM, LBW1122.

[62] Zachary Pousman, John Stasko, and Michael Mateas, 2007. Casual information visualization: Depictions of data in everyday life. *IEEE transactions on visualization and computer graphics 13*, 6, 1145-1152.

[63] Rebecca Raeside, Stephanie R Partridge, Anna Singleton, and Julie Redfern, 2019. Cardiovascular disease prevention in adolescents: eHealth, co-creation, and advocacy. *Medical Sciences 7*, 2, 34.

[64] Amon Rapp, Alessandro Marcengo, Luca Buriano, Giancarlo Ruffo, Mirko Lai, and Federica Cena, 2018. Designing a personal informatics system for users without experience in self-tracking: a case study. *Behaviour & Information Technology 37*, 4, 335-366.

[65] Verónica Rivera-Pelayo, Valentin Zacharias, Lars Müller, and Simone Braun, 2012. *Applying quantified self approaches to support reflective learning*. Association for Computing Machinery, Vancouver, British Columbia, Canada.

[66] John Rooksby, Mattias Rost, Alistair Morrison, and Matthew Chalmers Chalmers, 2014. Personal tracking as lived informatics. In *Proceedings of the 32nd annual ACM conference on Human factors in computing systems* ACM, 1163-1172.

[67] Judy Sebba, F Hunt, J Farlie, S Flowers, R Mulmi, and N Drew, 2009. Youth-led innovation: Enhancing the skills and capacity of the next generation of innovators.

[68] Tamar Sharon and Dorien Zandbergen, 2017. From data fetishism to quantifying selves: Self-tracking practices and the other values of data. *New Media & Society 19*, 11, 1695-1709.

[69] Steven & Polman Sommer, Joe, 2018. From Quantified Self to Building a More Fit Community; Data Tracking and Science Infographics as Boundary Objects. *Rethinking learning in the digital age: Making the Learning Sciences count, 13th International Conference of the Learning Sciences (ICLS) 2*, 1001-1004.

[70] Katta Spiel, Fares Kayali, Louise Horvath, Michael Penkler, Sabine Harrer, Miguel Sicart, and Jessica Hammer, 2018. Fitter, Happier, More Productive?: The Normative Ontology of Fitness Trackers. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems* ACM, alt08.

[71] Yulia Stepanchuk, 2017. Quantified Construction of Self: Numbers, Narratives and the Modern Individual. In *Proceedings of the International Conference IMS-2017* ACM, 28-36.

[72] Susanne Koch Stigberg, 2017. *A critical review on participation in mobile interaction design research*. Association for Computing Machinery, Stuttgart, Germany.

[73] Cristiano Storni, 2014. Design challenges for ubiquitous and personal computing in chronic disease care and patient empowerment: a case study rethinking diabetes self-monitoring. *Personal and Ubiquitous Computing 18*, 5, 1277-1290.

[74] Melanie Swan, 2009. Emerging patient-driven health care models: an examination of health social networks, consumer personalized medicine and quantified self-tracking. *International journal of environmental research and public health 6*, 2, 492-525.

[75] Nikolai Veresov, 2014. Refocusing the lens on development: Towards genetic research methodology. In *Visual methodologies and digital tools for researching with young children* Springer, 129-149.

[76] Lev Semenovich Vygotsky, 1997. *The collected works of LS Vygotsky: The history of the development of higher mental functions*. Springer Science & Business Media.

[77] Lev Semenovich Vygotsky, 1997. Educational psychology (R. Silverman, Trans.). *Boca Raton, FL: St. Lucie*, 173.

[78] Lev Semenovich Vygotsky, 1998. The collected works of LS Vygotsky, Vol. 5: Child psychology (MJ Hall, Trans; RW Reiber, Ed.) New York: Plenum, 55.

[79] Greg Walsh and Jennifer Golbeck, 2014. StepCity: a preliminary investigation of a personal informatics-based social game on behavior change. In *CHI'14 Extended Abstracts on Human Factors in Computing Systems* ACM, 2371-2376.

[80] Qinge Wu, Kelli Sum, and Dan Nathan-Roberts, 2016. How Fitness Trackers Facilitate Health Behavior Change. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting 60*, 1 (2016/09/01), 1068-1072. DOI= http://dx.doi.org/10.1177/1541931213601247.

[81] Hye Yang, Jae-Heon Kang, Ok Kim, Mona Choi, Myungju Oh, Jihyun Nam, and Eunju Sung, 2017. Interventions for preventing childhood obesity with smartphones and wearable device: a protocol for a non-randomized controlled trial. *International journal of environmental research and public health 14*, 2, 184.

[82] Robert K Yin, 2017. *Case study research and applications: Design and methods*. Sage publications.

[83] Jason C Yip, Elizabeth Foss, Elizabeth Bonsignore, Mona Leigh Guha, Leyla Norooz, Emily Rhodes, Brenna McNally, Panagis Papadatos, Evan Golub, and Allison Druin, 2013. Children initiating and leading cooperative inquiry sessions. In *Proceedings of the 12th International Conference on Interaction Design and Children* ACM, 293-296.